INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

ME 361
DYNAMICS OF MACHINERY

Lecture 5

FLYWHEELS AND GOVERNORS

Mr. Faisal Wahib Adam
Mechanical Engineering Department, KNUST

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Introduction

- Large variation in accelerations within mechanisms can cause significant variations in inertia and pin forces, supports reactions and torque required drive it at a constant or near constant speed. To stabilize the back-and-forth flow kinetic energy of rotating equipments, flywheels and governors are often attached to shafts.
- Flywheels are used in machines as an energy reservoir. They store energy during periods when the energy is more than required, and release energy to the system when the energy requirement is more than supplied.
- Governors are also used to control the mean speed of machines by regulating the flow working fluid, which power the machines. They rely on centrifugal force to function and control variation of speeds caused by load variation.

FLYWHEELS

Flywheels have the following functions:

- Reduce amplitude of speed fluctuation.
- Reduce maximum torque required.
- Stores and release energy when needed during cycling.





Speed Fluctuation

If ω_1 is the maximum speed of engine, ω_2 the minimum speed of engine, then the average speed, ω, is

$$\omega = \frac{\omega_1 + \omega_2}{2}$$

Coefficient of speed fluctuation, C_S, is defined as change in speed per average speed

$$C_S = \frac{\Delta\omega}{\omega} = \frac{\omega_1 - \omega_2}{\omega}$$

Another terminology which is sometimes used to describe speed fluctuation is coefficient of steadiness, defined as the inverse of Coefficient of speed fluctuation.

Energy Fluctuation

Coefficient of energy *fluctuation* is defined as the change in energy per work done in one cycle

$$C_E = \frac{\Delta E}{\text{Work done per Cycle}} = \frac{\Delta E}{\text{T}\theta}$$

where

θ is the angle of turn per cycle. The angles of turn for two-stroke and four-stroke engines are 2π and 4π respectively.

The average torque, T, may be defined in terms of power transmitted and angular speed as, $\text{T} = P/\omega$

where

P is the power developed by engine

ω is the average speed engine

Maximum Energy Fluctuation

The maximum fluctuation of energy is

$$\Delta E = E_1 - E_2 = \frac{1}{2} I\omega_1^{\,2} - \frac{1}{2} I\omega_2^{\,2} = \frac{1}{2} I(\omega_1^{\,2} - \omega_2^{\,2})$$

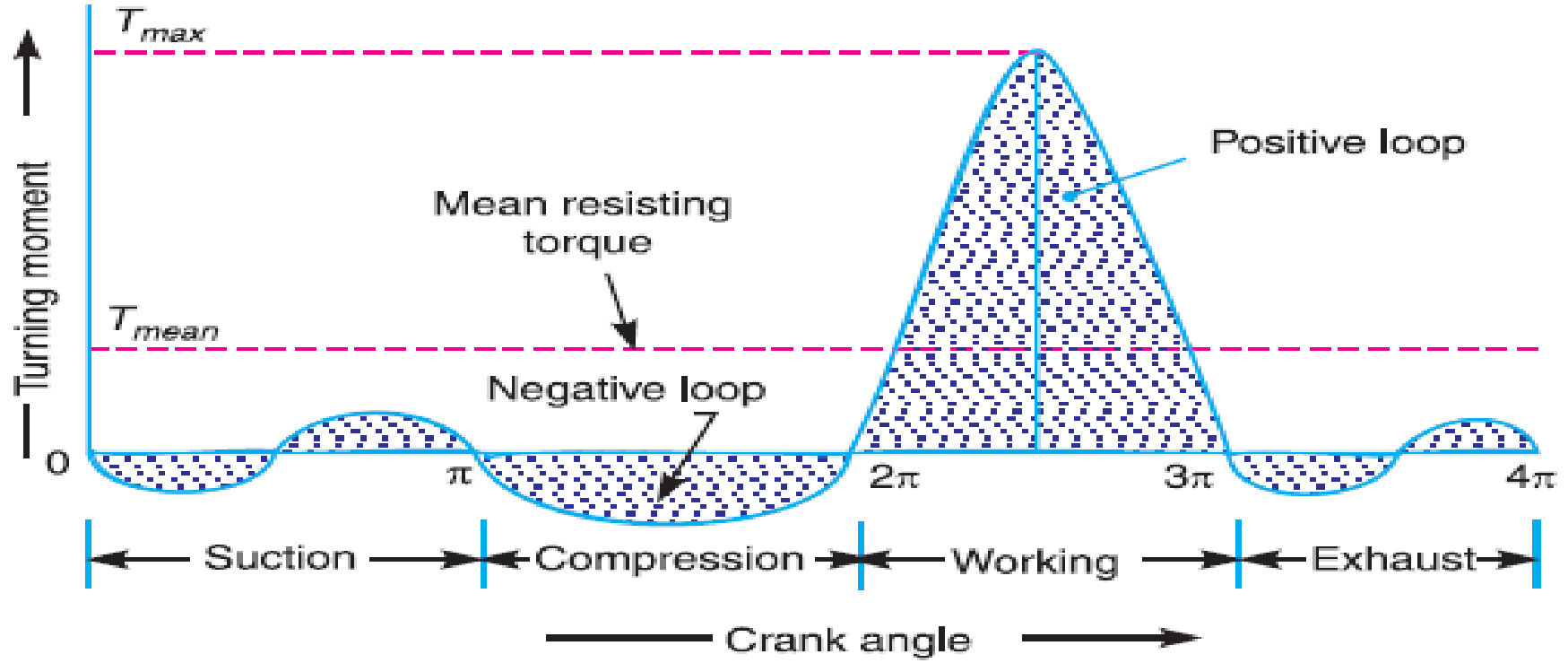
$$\Delta E = \frac{1}{2} I(\omega_1 + \omega_2)(\omega_1 - \omega_2) = I\omega(\omega C_S)$$

Therefore

$$\Delta E = I\omega^2 C_S$$

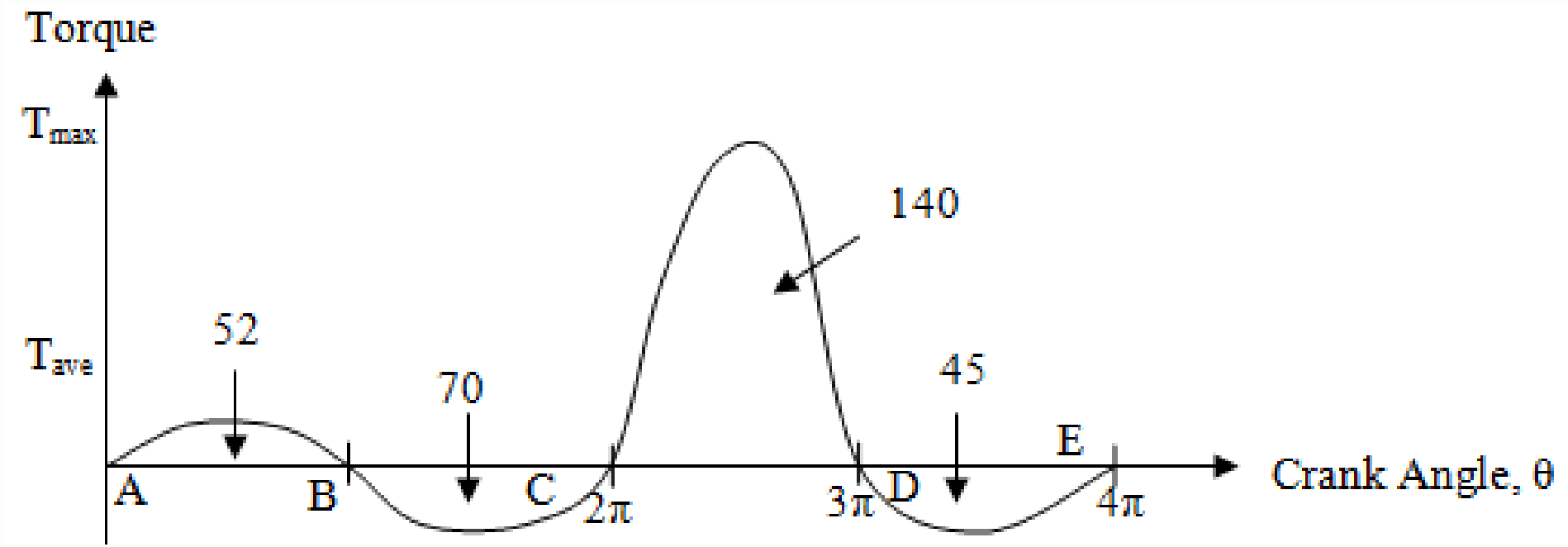
Turning Moment Diagram of One-Cylinder Four-Stroke IC Engine

Turning moment diagram (also known as crank effort diagram) is a plot of turning moment, T, on the vertical axis against the crank angle on the horizontal axis.



Example 6-1

The turning moment diagram for a two-stroke one-cylinder engine is shown below. If the mean torque is 77 N-m, find (a) the maximum energy fluctuation (b) coefficient of energy fluctuation.



Solution

Let the energy at A be E. Then, the energy at various points are:

Energy at B = E +52

Energy at C = E + 52-70 = E-18

Energy at D = E + 52-70+140 =E +122

Energy at E = E + 52-70+140-45 =E + 77

The maximum energy fluctuation is, ΔE = maximum of energy from A to E - maximum of energy from A to E

$\Delta E = (E + 122) - (E - 18) = 140.$

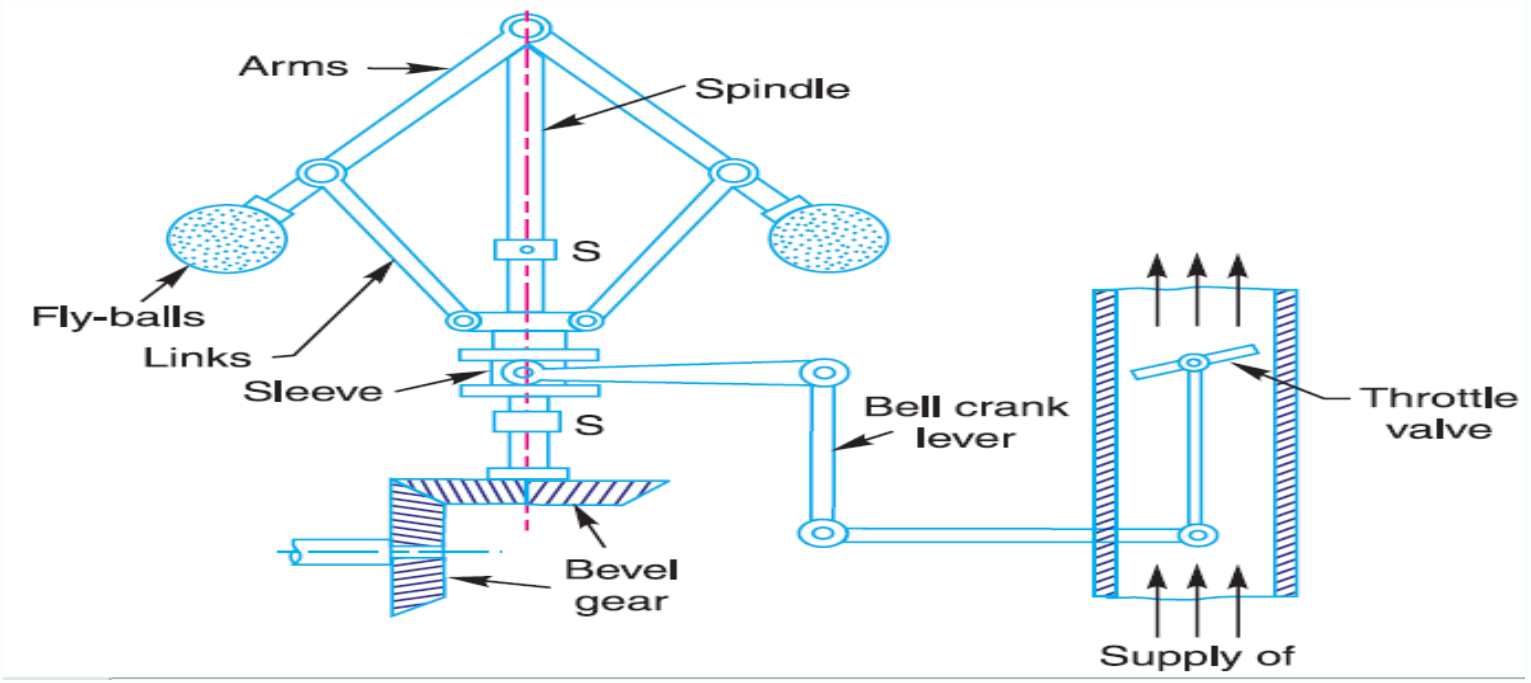
The coefficient of Energy fluctuation is given by

$$C_E = \frac{\Delta E}{\text{Work done per Cycle}} = \frac{\Delta E}{T\theta} = \frac{140}{77 \times 2\pi}$$

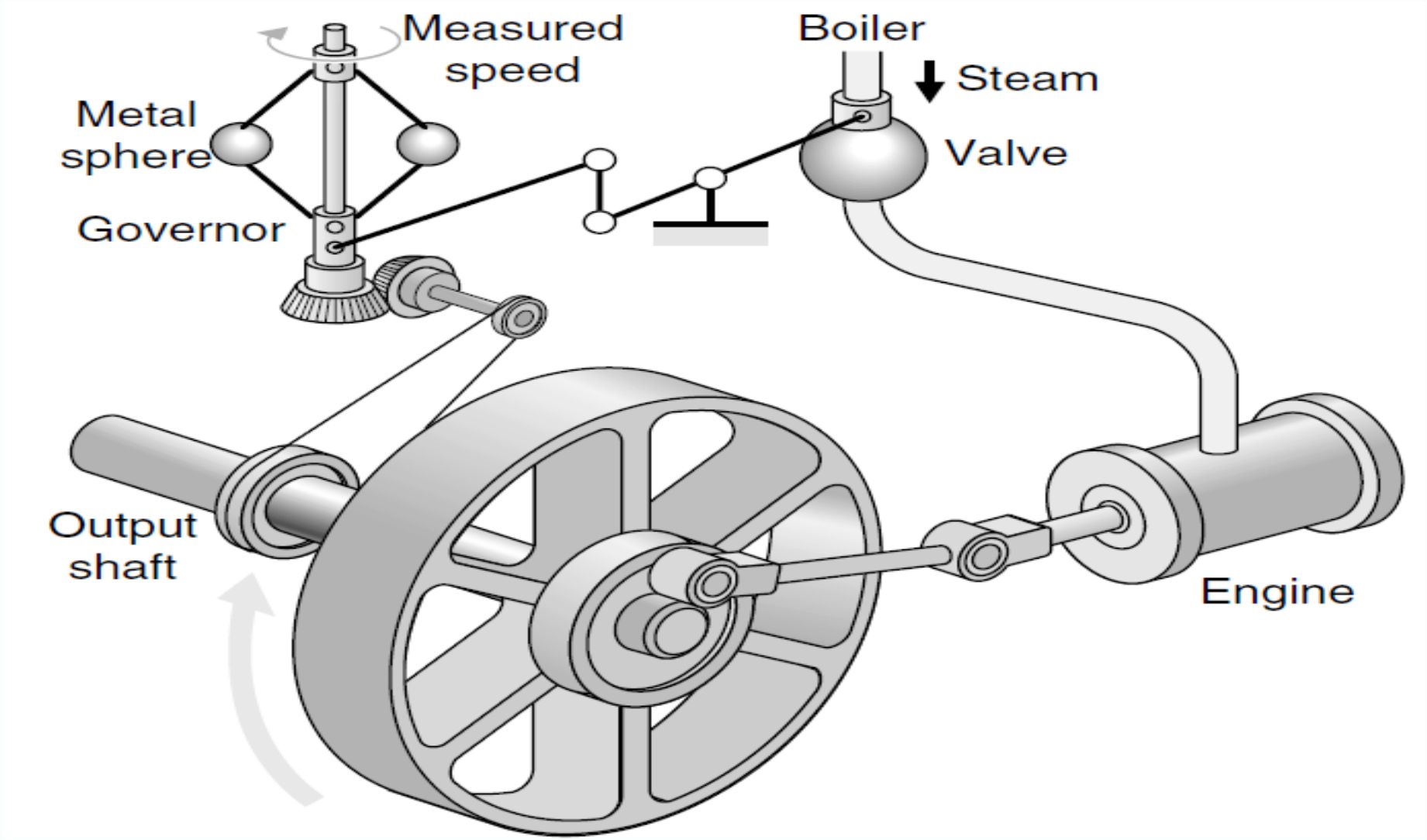
$$C_E = 0.289$$

GOVERNORS

Governors are broadly classified into centrifugal and inertia governors. Centrifugal governors are based on balancing of centrifugal force and moments of a rotating mass by an equal force and moments provided by either by action of a weight or springs.

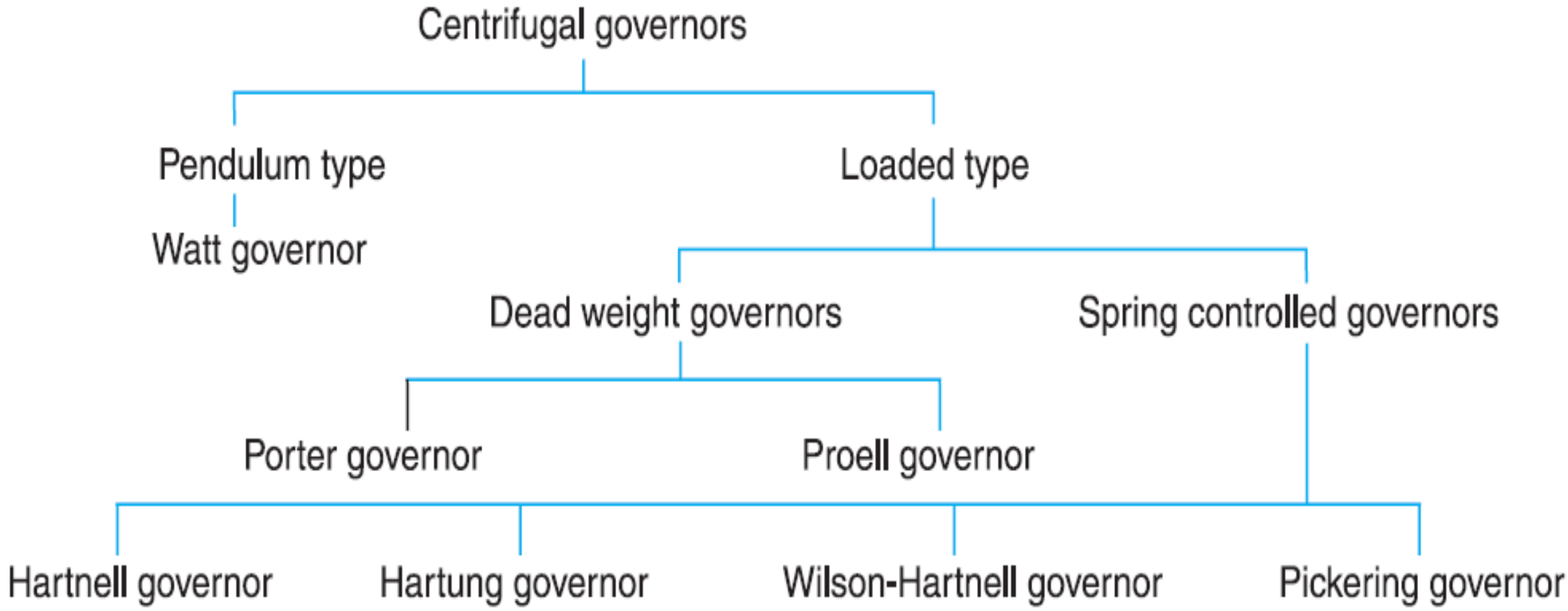


GOVERNORS

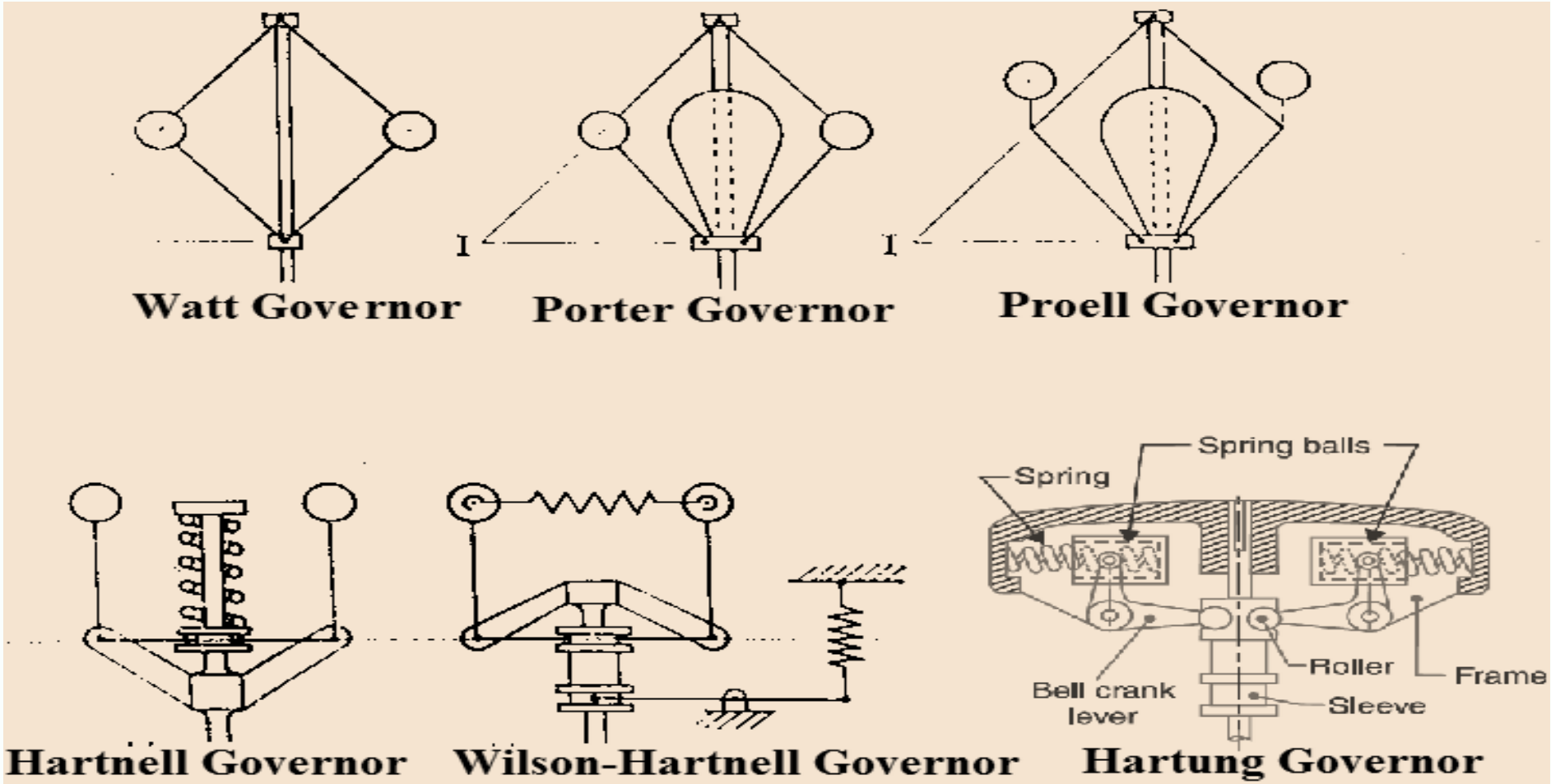

Measured
speed
Boiler
Steam
Metal
sphere
Valve
Governor
Output
shaft
Engine

Centrifugal Governors

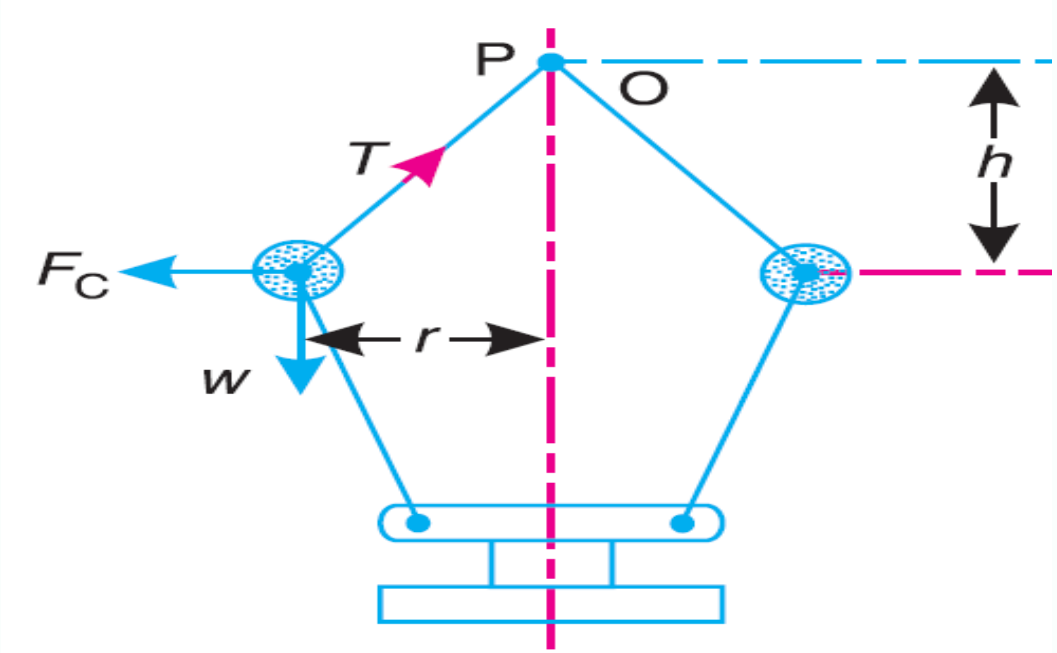
The centrifugal governors may further be classified as follows

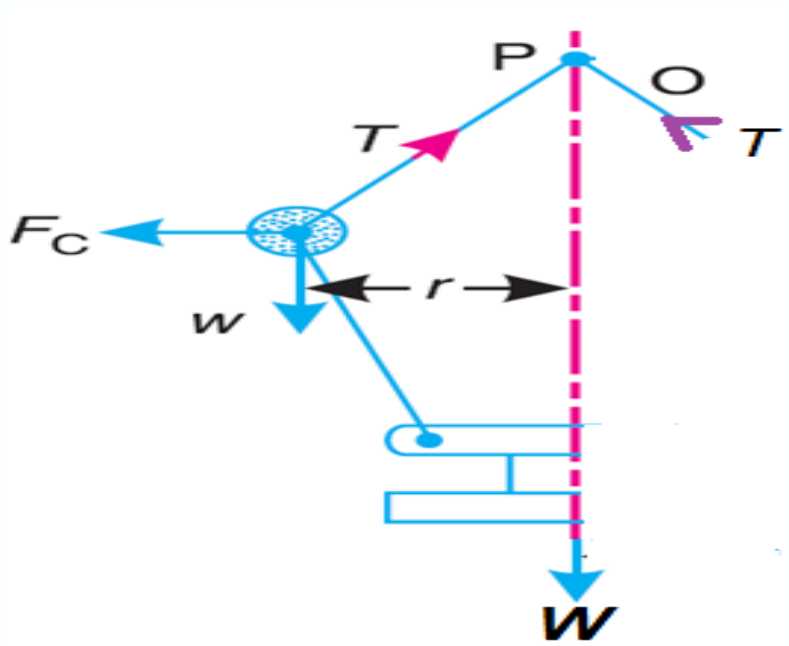


Centrifugal Governors



Watt Governor





$$\sum M_O = 0 \qquad \mathrm{F_C} \times h - w \times r = 0$$

F_C is the centrifugal force, $F_C = m\omega^2 r$

$$m\omega^2 rh = mgr$$

$$\therefore \omega^2 = g / h$$

Example

Find the vertical height of a Watt governor rotating at 65 rpm. Also determine the change in height corresponding to 1 rpm increase in the speed.

Solution:

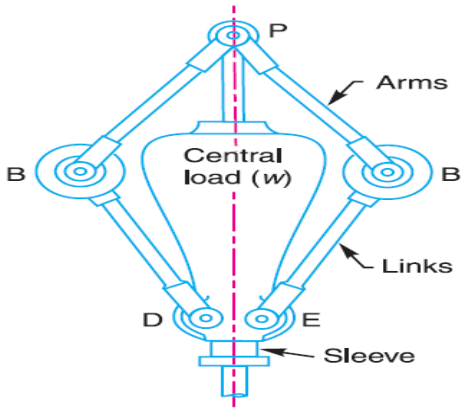
$$h = \frac{g}{\omega^2} = \frac{9.81}{\left(65\, x\, \frac{2\pi}{60}\right)^2} = 212\ mm$$

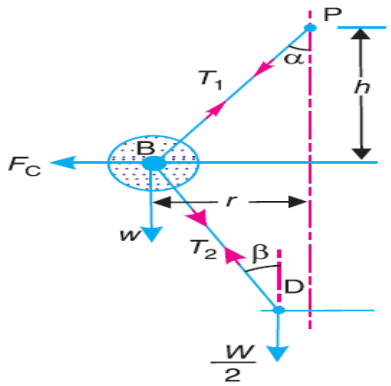
$$Partially\ differentiating\ \ h = \frac{g}{\omega^2}\ yields\ \ \ dh = -\frac{2g}{\omega^3}\, d\omega$$

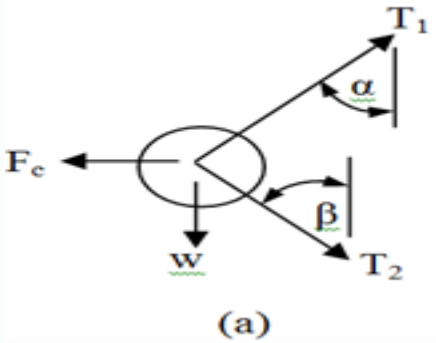
$$Substituting\ \omega = 65\ rpm\ and\ \ \ d\omega = 1\ rpm\ into\ the\ above\ equation\ gives$$

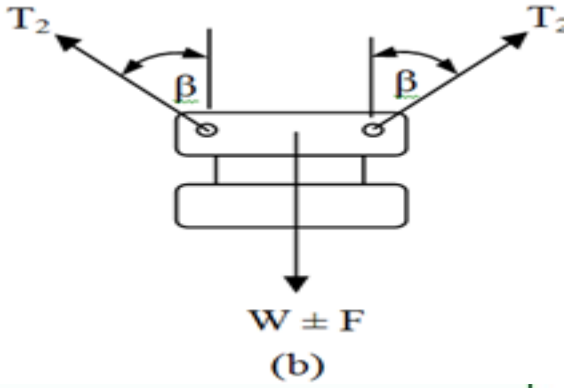
$$dh = -\frac{2g}{\omega^3}\, d\omega = -\frac{2(9.81)}{\left(65\, x\, \frac{2\pi}{60}\right)^3}\left(1\, x\, \frac{2\pi}{60}\right) = -6.515\ \ mm$$

Porter Governor









From (a)

$\sum F_x = 0; \quad F_C = T_1 sin\alpha + T_2 sin\beta \ldots\ldots\ldots 1$

$\sum F_y = 0; \quad w + T_2 cos\beta = T_1 cos\alpha \ldots\ldots\ldots 2$

From (b)

$\sum F_y = 0; \quad (W \pm F) = nT_2 cos\beta \ldots\ldots\ldots 3$

(3) into (1)

$$T_2 = \frac{W \pm F}{n cos\beta}$$

$$T_1 = \frac{w + (W \pm F)/n}{cos\alpha}$$

Substituting these into (1)

$$F_C = \left(\frac{w + (W \pm F)/n}{cos\alpha}\right) sin\alpha + \frac{W \pm F}{n cos\beta} sin\beta$$

$$F_C = \left(w + \frac{(W \pm F)}{n}\right) tan\alpha + \frac{(W \pm F)}{n} tan\beta$$

$$\frac{F_C}{tan\alpha} = w + \frac{(W \pm F)}{n}(1 + q)$$

$$where\ q = \frac{tan\beta}{tan\alpha}$$

Porter Governor

$$\omega^2 = \left[\frac{g}{r} + \frac{(W \pm F)}{nmr}(1+q)\right] tan\alpha$$

where $q = \frac{tan\beta}{tan\alpha}$

n= number of balls

m= mass of each ball

g= acceleration due to gravity

r= ball radius

W= weight of sleeve

F= frictional force

Example

The upper and lower arms of a three-ball Porter governor are 300 mm and 250 mm long, respectively. The upper arms are pivoted on the axis of rotation, and the lower arms pivoted on the sleeve at a distance 40 mm from the axis of rotation. The total mass of the sleeve and the load are 40 kg, and mass of each fly ball is 2 kg. Neglecting friction, determine the speed of rotation of the governor if the radius of each ball is 200 mm.

Solution

$$\alpha = \sin^{-1}\frac{200}{300} = 41.8^o; \beta = \sin^{-1}\frac{160}{250} = 39.8^o$$

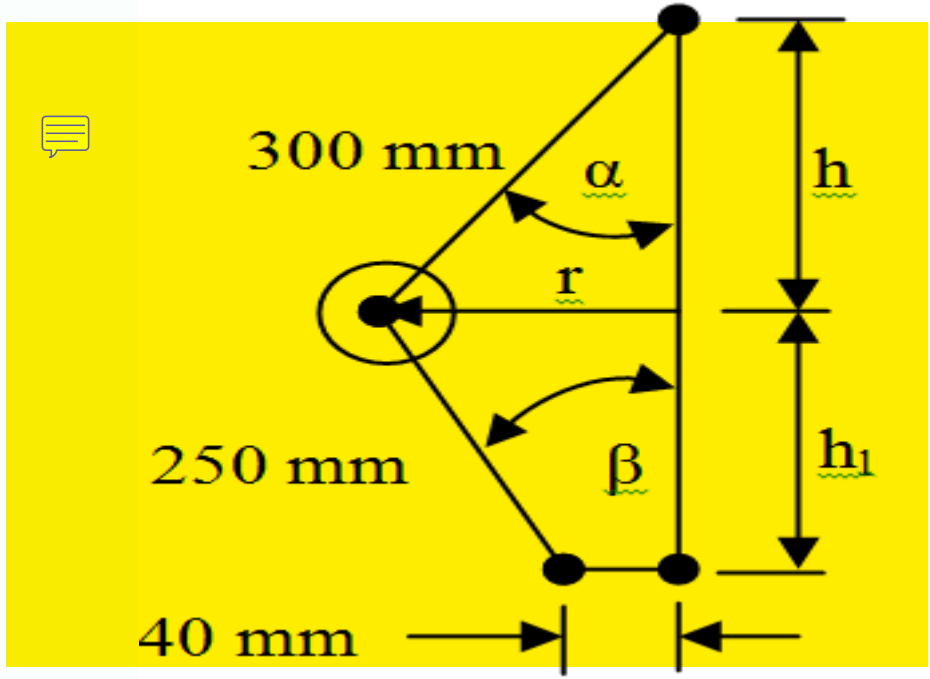
$$q = \frac{tan\beta}{tan\alpha} = 0.9318; \quad tan\alpha = 0.894$$

$$F = 0; n = 3; W = Mg = 392.4\ \text{N}; \text{w} = \text{mg} = 19.62\ \text{N}$$

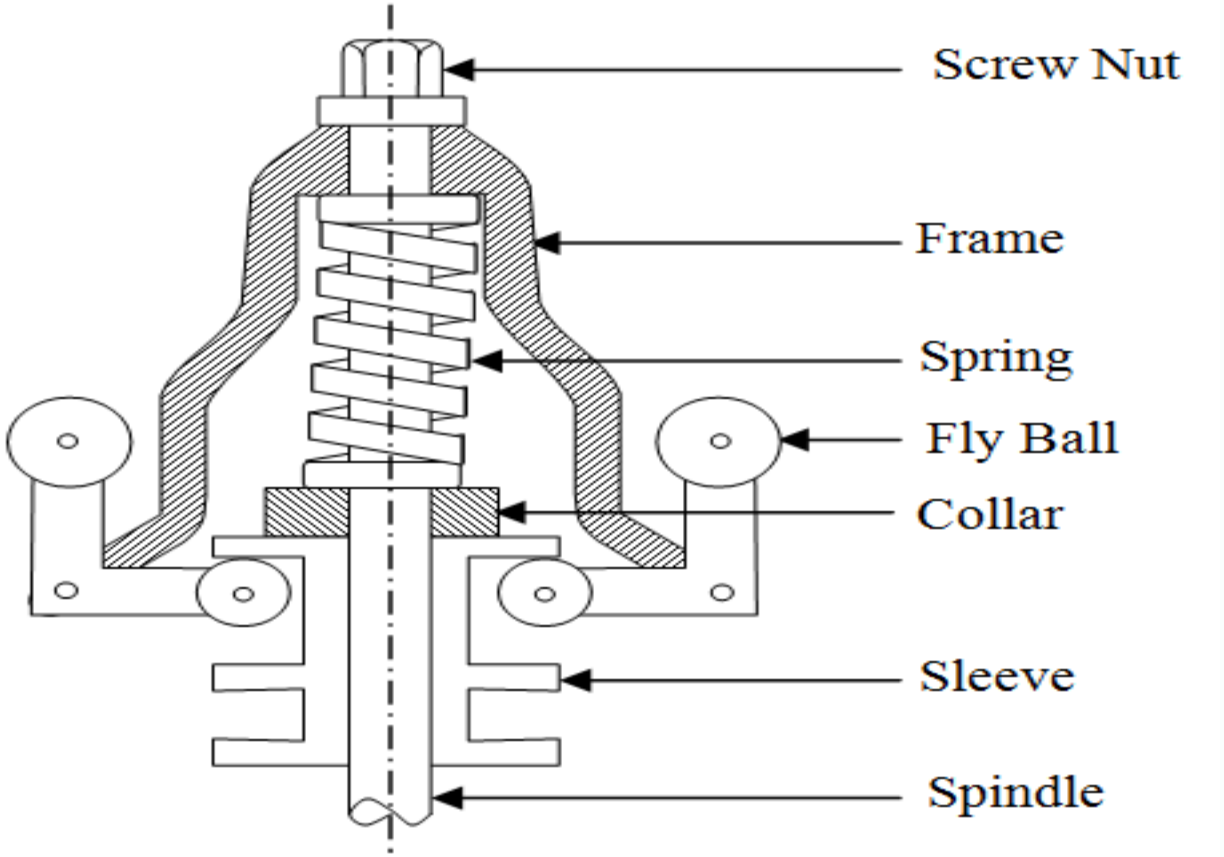
$$\frac{mr\omega^2}{tan\alpha} = w + \frac{(W \pm F)}{n}(1+q)$$

$$\frac{2\omega^2 \times 0.2}{0.894} = 19.62 + \frac{(392.4 \pm 0)}{3}(1 + 0.9318)$$

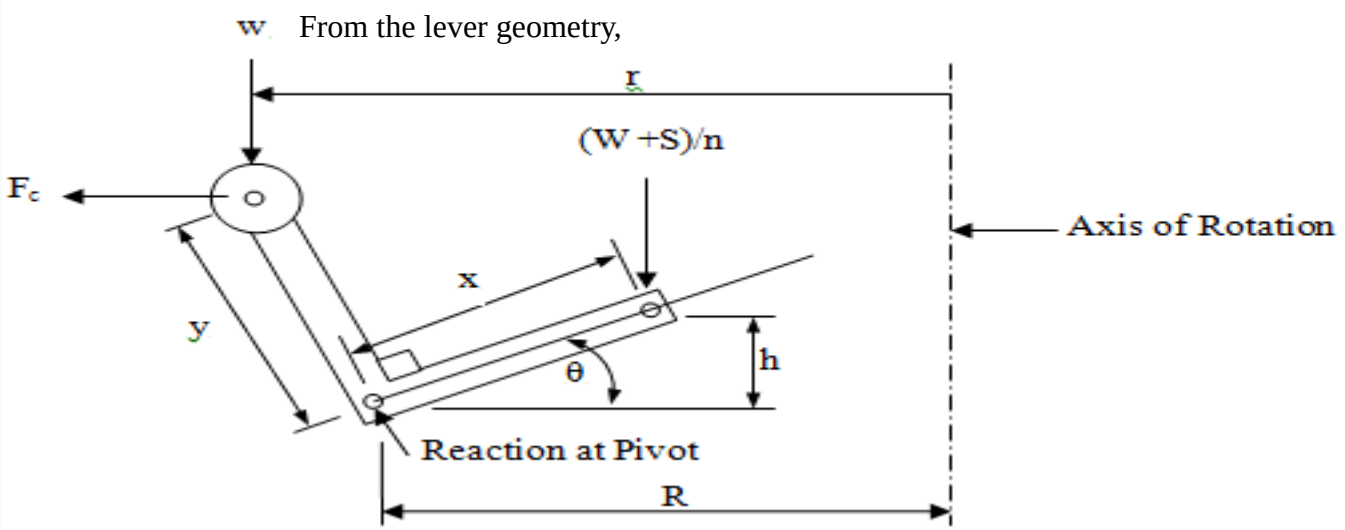
$$\omega = 24.67\ \text{rad/s} \quad \textit{or } 235.6\ \textit{rpm}$$



Hartnell Governor



Hartnell Governor



From the lever geometry,

$$\sum M_{pivot} = 0$$

$$F_C y\cos\theta + wy\sin\theta = \frac{1}{n}(W+S)x\cos\theta$$

$$F_C = m\omega^2 r = \frac{1}{n}(W+S)\frac{x}{y} - w\tan\theta$$

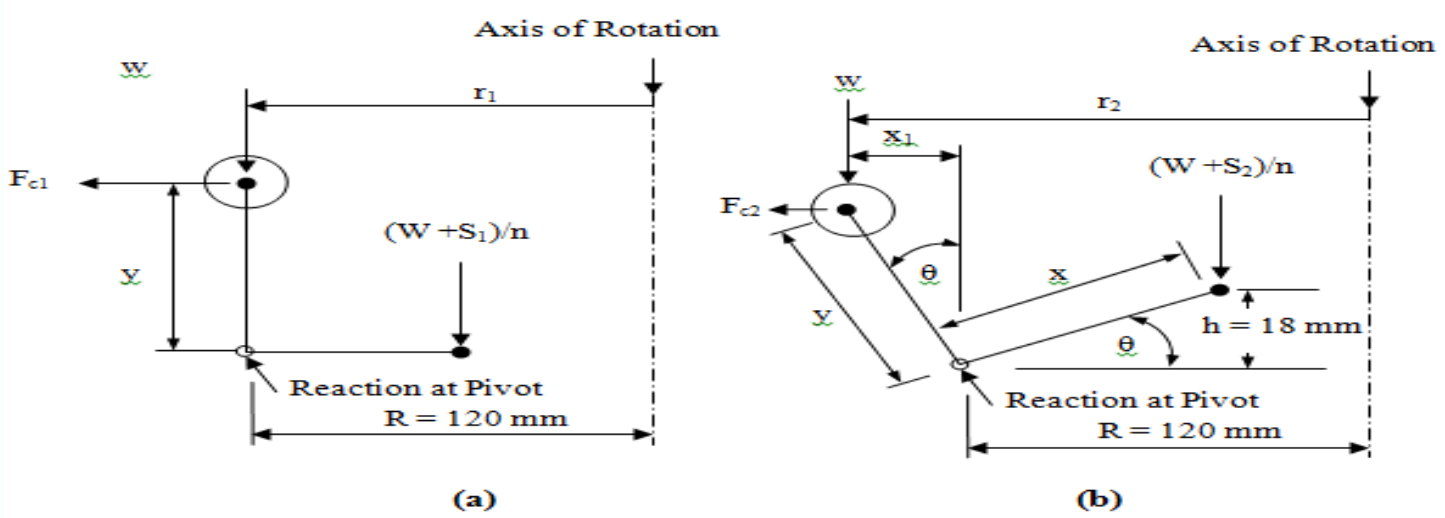
$$\omega^2 = \frac{1}{nmr}(W+S)\frac{x}{y} - \frac{g}{r}\tan\theta$$

From the lever geometry

$$r = R + y\sin\theta$$

Example

The extreme radii of rotations of fly balls of Hartnell type of governor are 80 mm and 120 mm. The governor has a central sleeve spring and three right-angled bell crank levers, which vertical and horizontal arms 120 mm and 80 mm, respectively. The levers are pivoted 120 mm from the axis of rotation. Each fly ball has a mass of 3 kg. Knowing that the two extreme speeds of the govern are 400 rpm and 450 rpm for a sleeve lift of 18 mm, and that the horizontal arm is perpendicular to the axis of rotation at the minimum speed, determine (a) total load on the sleeve at the lowest and highest speeds, (b) stiffness of the spring, and (c) deflection of the spring at the minimum speed if the mass of the sleeve is 50 kg. Neglect friction at all contacts.



Minimum, and (b) Maximum Positions

Solution cont'd…

$$(W + S_2) - (W + S_1) = S_2 - S_1 = 4498.3 - 2842.4$$

$$S_2 - S_1 = 1655.9 \text{ N}$$

$$S_2 - S_1 = kh = 1655.9$$

$$k = \frac{1655.9}{h} = \frac{1655.9}{18 \text{ x } 10^{-3}}$$

$$k = 92.0 \text{ kN/m}$$

$$S_1 = 1655.9 - W = 1655.9 - 50 \text{ x } 9.81$$

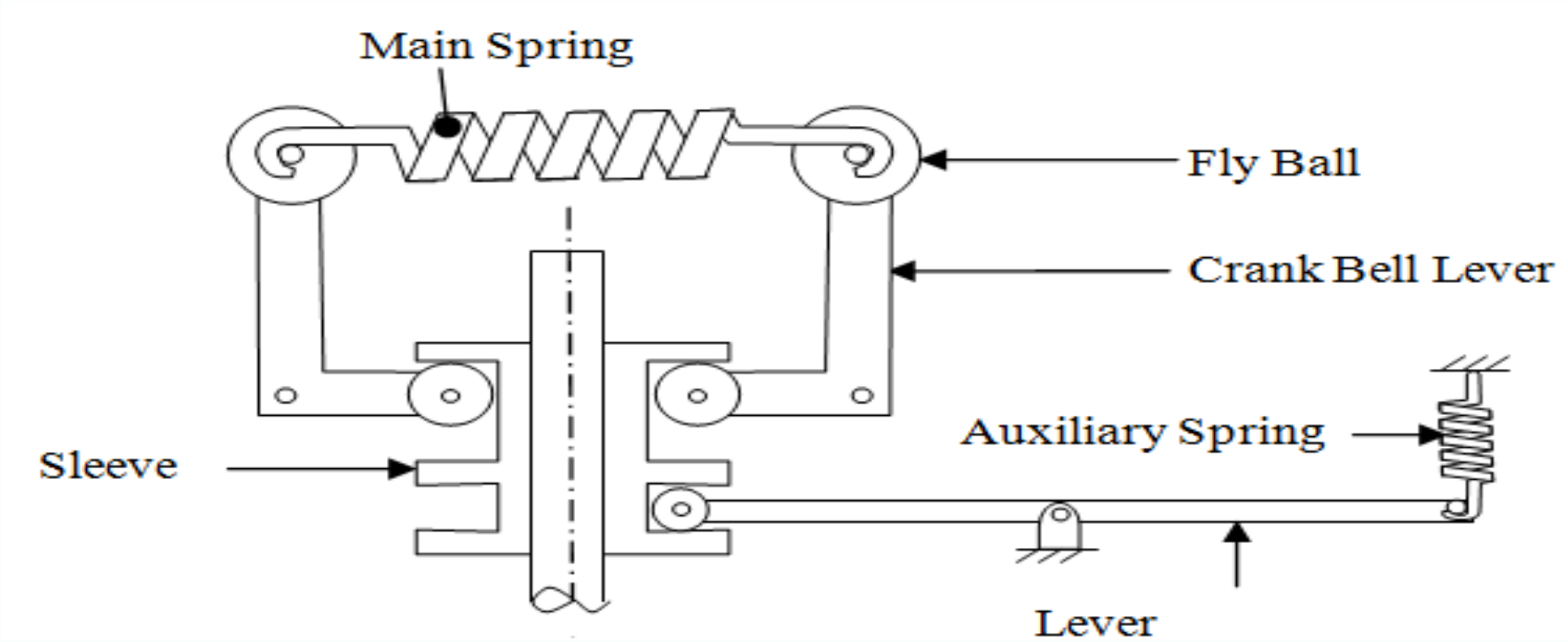
$$S_1 = 1165.4 \text{ N}$$

$$S_1 = kh_0$$

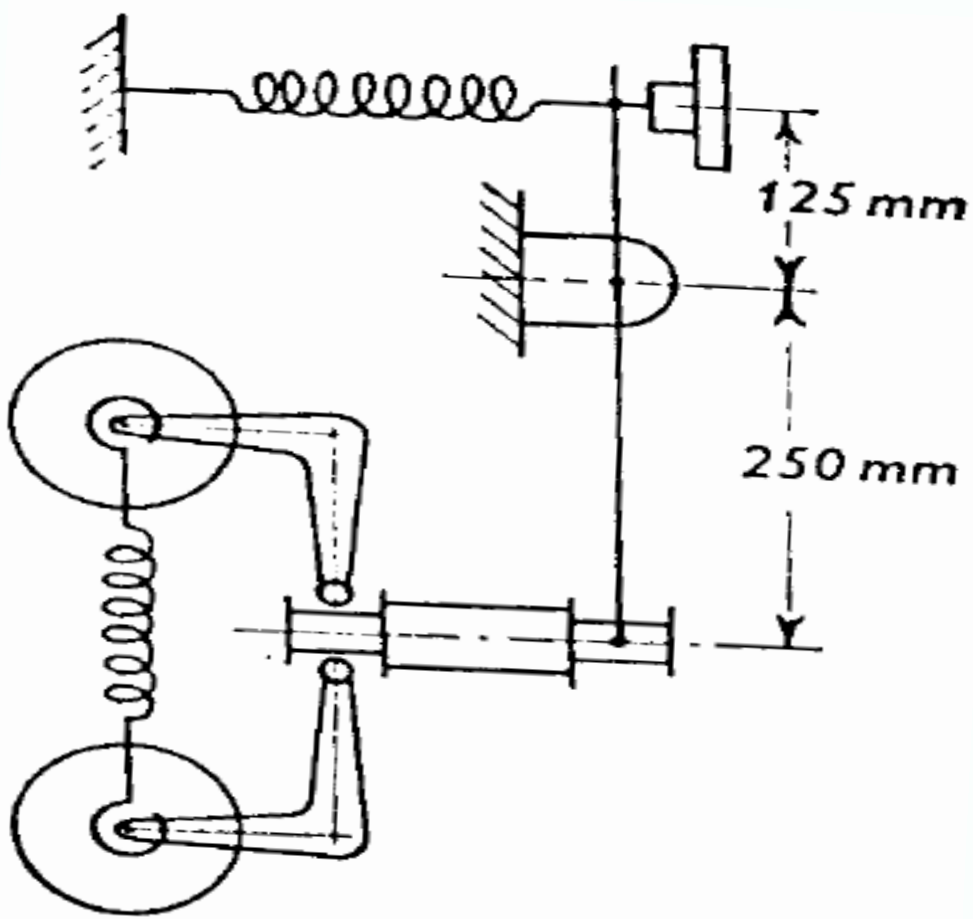
$$h_0 = \frac{S_1}{k} = \frac{1165.4}{92.0 \text{ x } 10^3}$$

$$h_0 = 0.0196 \text{ m}$$

Wilson-Hartnell governors and Auxiliary Control Force Elements



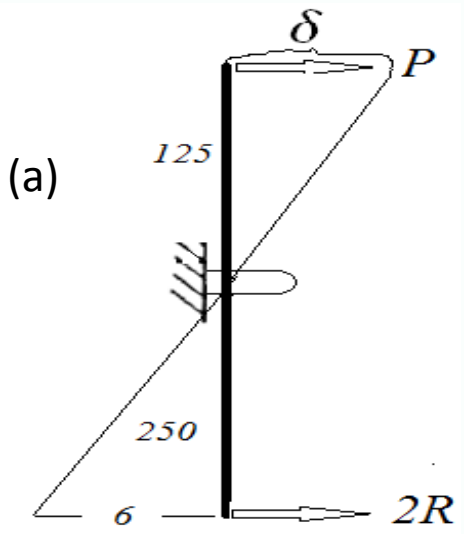
Example

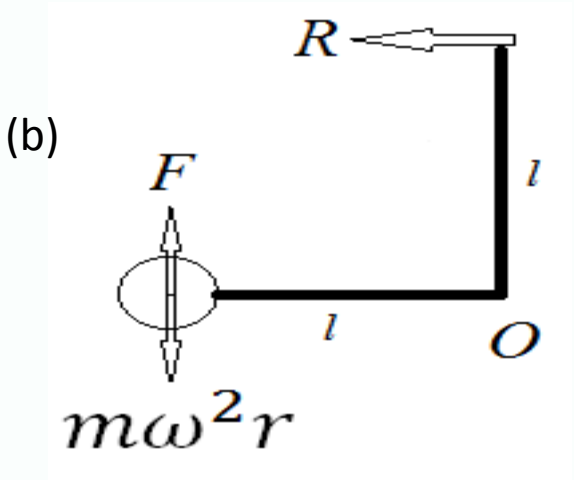


The figure shows a throttle valve governor. Each ball has a mass of 6.5 kg and the arms are of equal length. The speed range is 425 to 440 rev/min with a sleeve movement of 6 mm. The minimum ball path radius is 122 mm. If the combined strength of the ba11 springs is 15 kN/m of extension, find the rate of the auxiliary spring.

Solution

FBD





From (a), $\sum M_0 = 0$

125P-2R(250)=0; R=0.25P

From (b), $\sum M_0 = 0$

$$F - m\omega^2 r = R = 0.25P$$

F is the initial tension in the ball spring

Therefore

@top speed

$$F - 6.5\left(425 \times \frac{2\pi}{60}\right)^2 \times 0.122 = 0.25P_1$$

$$F - 1570.75 = 0.25P_1 \ldots\ldots\ldots 1$$

@Bottom speed

$$(F + 180) - 6.5\left(440 \times \frac{2\pi}{60}\right)^2 \times (0.122 + 0.006) = 0.25P_1$$

$$F - 1586.385 = 0.25P_2 \ldots\ldots\ldots 2$$

$$k = \frac{\Delta P}{\Delta \delta} = \frac{62.549}{3 \times 10^{-3}} = 20.85\ KN/m$$

Properties of Governors

The **Effort** of a governor is the force exerted at the sleeve for a given fractional change in speed.

The **Power** of a governor is the work done at the sleeve for a given fractional change in speed, i.e. power= mean effort x sleeve movement

Sensitiveness(coefficient of fluctuation of speed)

The sensitiveness of a governor is defined as the ratio of the difference between the maximum and minimum equilibrium speeds to the mean speed.

Hunting

When a governor is too sensitive, the governor causes the speed of the engine or the system is controlling to fluctuate continuously above and below the equilibrium speed. Such a governor is said to be hunting.

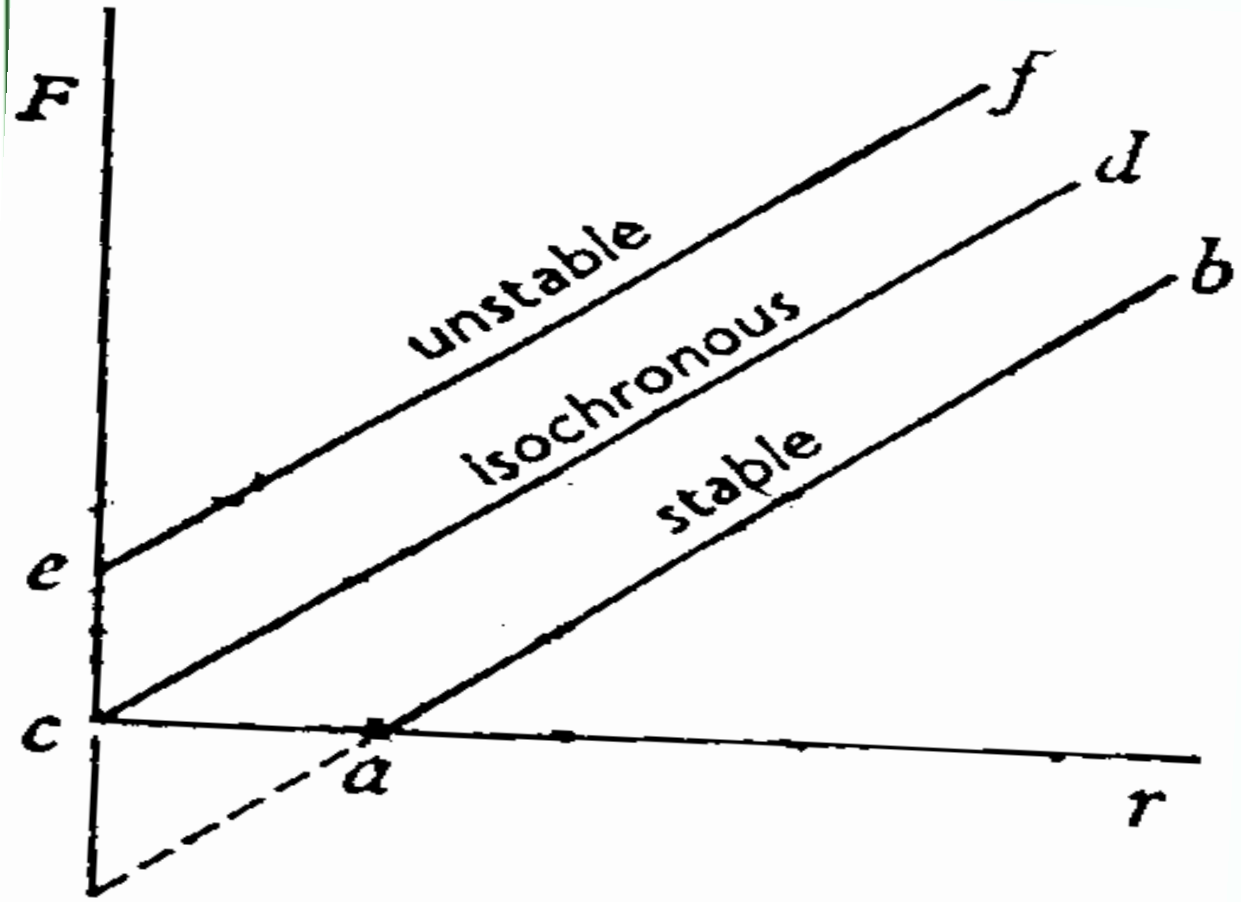
Stability

A governor is said to be stable there is only one radius of rotation of the fly balls for a particular speed with the operating range at which the governor is in equilibrium. For stable governor, the radius of rotation increases with the speed increases.

Isochronous

A governor is said to be isochronous when the equilibrium speed is constant for all radii of rotation of fly balls with the working range. Gravity controlled governors such as Porter governor are not isochronous, but spring loaded governors are capable. The isochronous governors have no practical use because the sleeve move to its extreme position immediately the speed deviates from the isochronous speed. It is possible to use this governor as angular speed-dependent toggle position switch.

Properties of Governors



For **stable governor**, all radius of rotation increases with the speed increase.

$$F = ar - e;\ \frac{F}{r} = a - \frac{e}{r}F$$

$$\frac{dF}{dr} = a;\ \therefore\ \frac{dF}{dr} > \frac{F}{r}$$

Isochronous

A governor is said to be isochronous when the equilibrium speed is constant for all radii of rotation.

$$F = ar;\ \frac{F}{r} = a - \frac{e}{r}$$

$$\frac{dF}{dr} = a;\ \frac{dF}{dr} = \frac{F}{r}$$

Unstable

$$F = ar + e;\ \frac{F}{r} = a + \frac{e}{r}$$

$$\frac{dF}{dr} = a;\ \frac{dF}{dr} < \frac{F}{r}$$

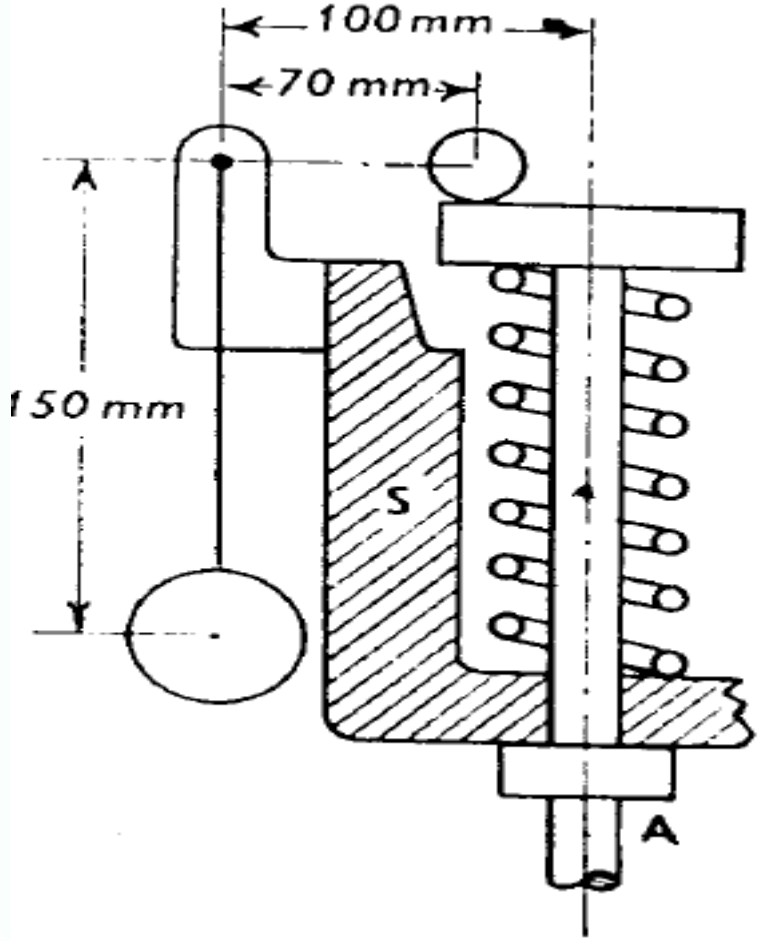
Assignments 1

1. The turning moment diagram for a multi cylinder engine has been drawn to a scale of 1 mm = 4500 N -m vertically and 1 mm = 2.4 0 horizontally. The intercepted areas between output torque curve and mean resistance line taken in order from one end are 342, 23, 245, 303, 115, 232, 227, 164 mm^2, when the engine is running at 150 rpm. If the mass of the flywheel is 1000 kg and the total fluctuation of speed does not exceed 3% of the mean speed, find the minimum value of the radius of gyration.

Assignment 2

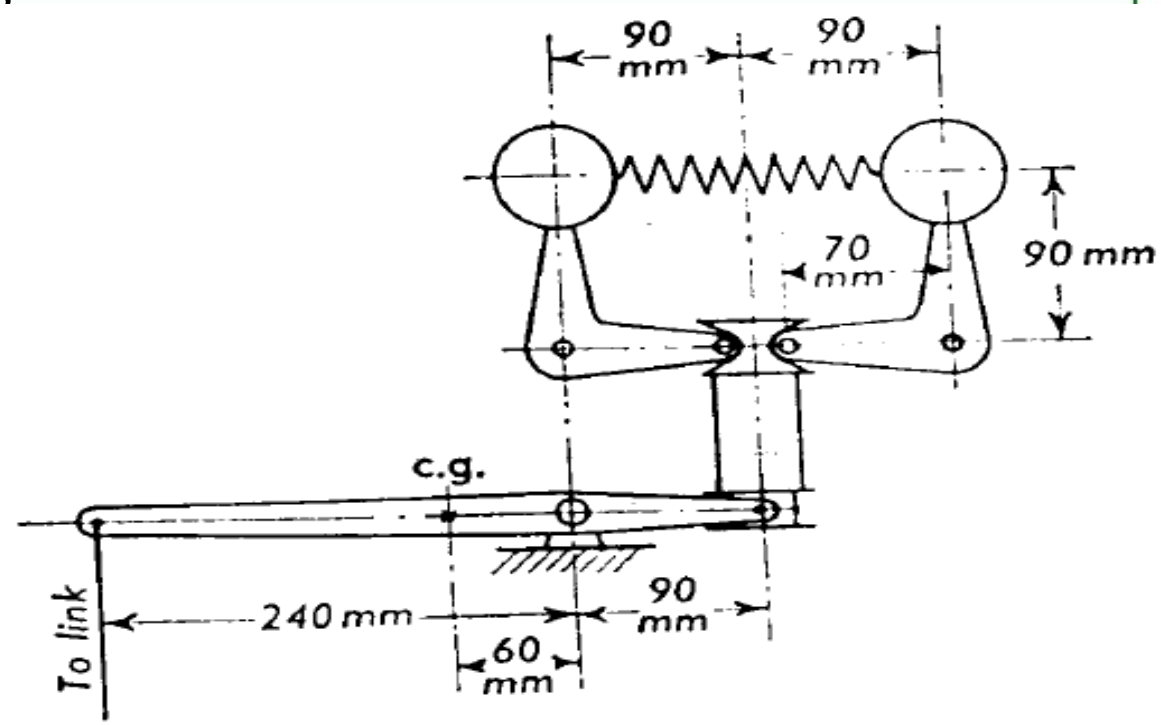
The figure shows the arrangement of a governor. The central sleeve A does not move axially. The sleeve S has a mass of 18 kg and the frictional resistance to its movement may-be taken as 20 N. There are two right-angled bell-crank levers, each having a ball of mass 4 kg attached to its lower end. The masses of the levers themselves and the rollers on their horizontal arms may be neglected but the dead weight of the balls must not *be* neglected. The spring has a stiffness of 50 kN/m compression. When the sleeve is in its lowest position, the balls rotate in a circle of 100 mm radius, the ball arms are vertical and the spring force is 550 N. Calculate *(a)* the speed at which the sleeve will begin to rise from its lowest position, and

(b) the range within which the speed must lie when the sleeve is 10 mm above its lowest position.



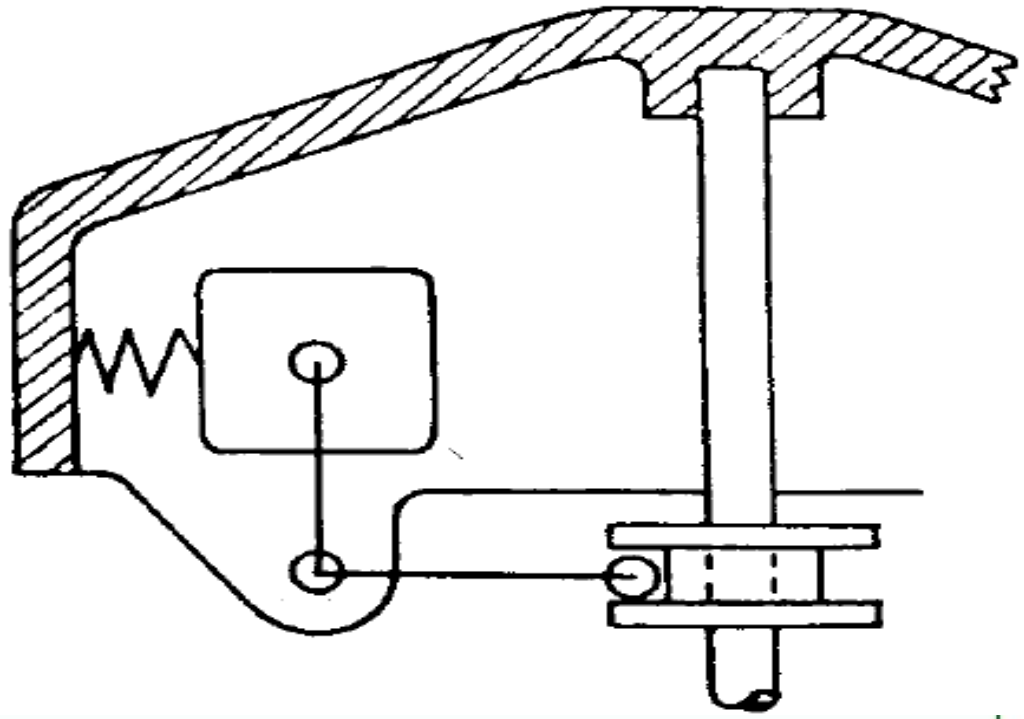
Assignments 3

A governor with the control spring directly connected between the masses operates a hanging link through a lever as shown. For the positior given, the particulars are as follows: spring tensior 650 N; elastic force of spring 11 kN/m; mass o spring 2.2 kg; mass of each ball 2.7 kg; mass o each bell-crank 0.9 kg; radius of gyration of bell crank about fulcrum 30 mm; mass of sleeve 1.6 kg mass of lever 2.7 kg; c.g. of lever as marked; radius of gyration of lever about fulcrum 70 mm; mass o hanging link 3.6 kg. Determine the governor speed for the position shown, neglecting the contro effects of the bell-cranks,



Assignment 4

The bell-crank levers of a spring-controlled governor are carried by It. rotating casing, as shown. Each ball arm is 100 mm long, each sleeve arm is125 mm long and when the sleeve is in its mid-position, the ball arms are vertical, the sleeve arms are horizontal and the radius of the ball path is 150 mm. The balls have a mass of 3.6 kg each and arc controlled by helical springs in initial compression in contact with the outside of the balls and the inside of the casing. The operating gear-reduced to the sleeve-has a mass of 9 kg. Each of the springs has stiffness of 10 kN/m and is initially compressed 50 mm for the bottom position of the sleeve. Determine the range of speed of operation of the governor for a sleeve lift of 25 mm.



THANK YOU